

Generic Type Instantiation

Example 1 : Basic Type Instantiation

Apply type parameters to polymorphic types using square brackets:

$\text{seq } \mathbb{N}$
 $\mathbb{P } X$
 $\emptyset[\mathbb{Z}]$

These instantiate generic types for specific type arguments.

Example 2 : Empty Set Instantiation

$\text{emptyNats} : \mathbb{P } \mathbb{N}$ $\text{emptyInts} : \mathbb{P } \mathbb{Z}$
$\text{emptyNats} = \emptyset[\mathbb{N}]$ $\text{emptyInts} = \emptyset[\mathbb{Z}]$

emptyset is generic; $\text{emptyset}[\mathbb{N}]$ is the empty set of naturals.

Example 3 : Sequence Type Instantiation

$\text{numbers} : \text{seq } \mathbb{N}$ $\text{integers} : \text{seq } \mathbb{Z}$
$\text{numbers} = \langle 1, 2, 3 \rangle$ $\text{integers} = \langle -1, 0, 1 \rangle$

$\text{seq}[\mathbb{N}]$ is the type of sequences of natural numbers.

Example 4 : Power Set Instantiation

$\text{subsetsOfN} : \mathbb{P}(\mathbb{P } \mathbb{N})$ $\text{example} : \mathbb{P}(\mathbb{P } \mathbb{N})$
$\text{example} = \{\{1, 2\}, \{3\}, \emptyset[\mathbb{N}]\}$

$\mathbb{P}(\mathbb{P } \mathbb{N})$ is the power set of the power set of naturals (sets of sets).

Example 5 : Complex Type Parameters

$\text{pairsSeq} : \text{seq}(\mathbb{N} \times \mathbb{N})$ $\text{pairsSet} : \mathbb{P}(\mathbb{N} \times \mathbb{N})$
$\text{pairsSeq} = \langle (1, 2), (3, 4) \rangle$ $\text{pairsSet} = \{(1, 2), (3, 4)\}$

Type parameters can be complex types like $\mathbb{N}$ cross $\mathbb{N}$.

Example 6 : Nested Instantiation

$\text{seqOfSets} : \text{seq}(\mathbb{P } \mathbb{N})$ $\text{setOfSeqs} : \mathbb{P}(\text{seq } \mathbb{N})$
$\text{seqOfSets} = \langle \{1, 2\}, \{3, 4, 5\} \rangle$ $\text{setOfSeqs} = \{\langle 1, 2 \rangle, \langle 3, 4 \rangle\}$

$\text{seq}[\mathbb{P } \mathbb{N}]$ is a sequence of sets. $\mathbb{P}[\text{seq}[\mathbb{N}]]$ is a set of sequences.

Example 7 : Generic Function Instantiation

$[X]$	$\text{identity} : X \rightarrow X$
	$\forall x : X \bullet \text{identity}(x) = x$

$\text{id_nat} : \mathbb{N} \rightarrow \mathbb{N}$ $\text{id_int} : \mathbb{Z} \rightarrow \mathbb{Z}$
$\text{id_nat} = \text{identity}[\mathbb{N}]$ $\text{id_int} = \text{identity}[\mathbb{Z}]$

$\text{identity}[\mathbb{N}]$ instantiates the generic identity function for natural numbers.

Example 8 : Multiple Type Parameters

$[X, Y]$	$\text{pairType} : X \times Y$
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$\text{natIntPair} : \mathbb{N} \times \mathbb{Z}$ $\text{intNatPair} : \mathbb{Z} \times \mathbb{N}$
$\text{natIntPair} = \text{pairType}[\mathbb{N}, \mathbb{Z}]$ $\text{intNatPair} = \text{pairType}[\mathbb{Z}, \mathbb{N}]$

Instantiate with multiple type arguments.

Example 9 : Bag Instantiation

$\text{natBag} : \text{bag } \mathbb{N}$ $\text{intBag} : \text{bag } \mathbb{Z}$
$\text{natBag} = \llbracket 1, 1, 2, 3 \rrbracket$ $\text{intBag} = \llbracket -1, 0, 1 \rrbracket$

$\text{bag}[\mathbb{N}]$ is the type of bags (multisets) of natural numbers.

Example 10 : Generic Relation Types

$\text{natRelation} : \mathbb{N} \leftrightarrow \mathbb{N}$ $\text{mixedRelation} : \mathbb{N} \leftrightarrow \mathbb{Z}$
$\text{natRelation} = \{1 \mapsto 2, 2 \mapsto 4\}$ $\text{mixedRelation} = \{1 \mapsto -1, 2 \mapsto -2\}$

Relations between different types.

Example 11 : Sequence of Pairs

$\text{coordSeq} : \text{seq}(\mathbb{N} \times \mathbb{N})$
$\text{coordSeq} = \langle (0, 0), (1, 1), (2, 4), (3, 9) \rangle$

A sequence where each element is a pair (coordinate).

Example 12 : Set of Functions

$square : \mathbb{N} \rightarrow \mathbb{N}$
$double : \mathbb{N} \rightarrow \mathbb{N}$
$successor : \mathbb{N} \rightarrow \mathbb{N}$
$\forall n : \mathbb{N} \bullet square(n) = n * n$
$\forall n : \mathbb{N} \bullet double(n) = 2 * n$
$\forall n : \mathbb{N} \bullet successor(n) = n + 1$
$funcSet : \mathbb{P}(\mathbb{N} \rightarrow \mathbb{N})$
$funcSet = \{square, double, successor\}$

A set containing three functions.

Example 13 : Practical Example - Generic Pair

$[T]$
$makePair : T \times T \rightarrow T \times T$
$\forall x, y : T \bullet makePair(x, y) = (x, y)$

Generic pair function that works with any type T.

Example 14 : Generic Pair Extraction

$[X]$
$fst : X \times X \rightarrow X$
$snd : X \times X \rightarrow X$
$\forall x, y : X \bullet fst(x, y) = x$
$\forall x, y : X \bullet snd(x, y) = y$

Generic first and second functions that work with any type X.

Example 15 : List of Lists

$matrix : seq(seq \mathbb{N})$
$matrix = \langle \langle 1, 2, 3 \rangle, \langle 4, 5, 6 \rangle, \langle 7, 8, 9 \rangle \rangle$

Nested sequences represent a 3x3 matrix.

Example 16 : Homogeneous Collections

Type instantiation ensures homogeneity:

$validSeq : seq \mathbb{N}$
$validSeq = \langle 1, 2, 3 \rangle$

INVALID: $seq[\mathbb{N}]$ cannot contain non-naturals $invalidSeq = \langle 1, -1, 2 \rangle$ – would be type error if -1 not in $\mathbb{N}$

Example 17 : Generic Queue

$[T]$	
$wrap : T \rightarrow \text{seq } T$	
$unwrap : \text{seq } T \rightarrow T$	
$\forall x : T \bullet wrap(x) = \langle x \rangle$	
$\forall s : \text{seq } T \bullet \#s > 0 \Rightarrow unwrap(s) = s(1)$	

Generic wrap/unwrap functions for working with sequences.

Example 18 : Type Synonyms with Instantiation

Type abbreviations can use instantiated types:

IntSet == P Z (set of integers) NatSeq == seq N (sequence of naturals) Coordinate == N cross N (pair of naturals) Path == seq Coordinate (sequence of coordinates)

These provide readable names for complex instantiated types.

Example 19 : Higher - Order Types

$transformers : \mathbb{P}(\mathbb{N} \rightarrow \mathbb{N})$
$sequences : \mathbb{P}(\text{seq } \mathbb{N})$
$transformers = \{square, double\}$
$sequences = \{\langle 1, 2 \rangle, \langle 3, 4 \rangle\}$

Sets of functions and sequences.

Example 20 : Instantiation elem Quantifiers

$\forall s : \text{seq } \mathbb{N} \bullet \#s \geq 0$

$\exists p : \mathbb{N} \times \mathbb{N} \mid p \bullet 1 = p.2$

Quantifying over instantiated generic types.

Example 21 : Partial Function Types

$lookup : \mathbb{N} \mapsto \text{seq } \mathbb{N}$
$mapping : \mathbb{N} \leftrightarrow \mathbb{N}$
$lookup = \{1 \mapsto \langle 10, 20 \rangle, 2 \mapsto \langle 30, 40 \rangle\}$
$mapping = \{1 \mapsto 2, 2 \mapsto 4, 3 \mapsto 6\}$

Partial functions with instantiated types.

Example 22 : Best Practices

When using generic instantiation: 1. Be explicit about type parameters for clarity 2. Use consistent naming (seq[T], not mixed seq T and T seq) 3. Document what type parameters represent 4. Instantiate at the right abstraction level 5. Prefer generic definitions over duplicate specific definitions 6. Type-check that instantiations make sense

Example 23 : Common Patterns

Common instantiation patterns: - $\text{seq}[T]$ for sequences of T - $P[T]$ for sets of T - $T \rightarrowtail U$ for partial functions - $T \leftrightarrow U$ for relations - $\text{bag}[T]$ for multisets of T - $T \times U$ for pairs - $T \rightarrow U$ for total functions

Example 24 : Type Safety

Generic instantiation provides type safety:

$$\frac{\text{safe} : \text{seq } \mathbb{N}}{\text{safe} = \langle 1, 2, 3 \rangle}$$

Type checker ensures all elements match the instantiated type.

COMPILE ERROR examples: - $\text{seq}[\mathbb{N}] = \langle 1, 2, 'a' \rangle$ - $'a'$ is not a natural - $P[\mathbb{N}] = \{1, 2, -1\}$ - -1 might not be in $\mathbb{N}$ depending on $\mathbb{N}$ definition - $\text{bag}[\mathbb{Z}] = [[1, \text{"hello"}]]$ - "hello" is not an integer

The type system catches these errors at compile/check time.